Candidate's Name: …….....

Signature: ……...

Random No.						Personal No.		

(*Do not write your School / Center Name or Number anywhere on this booklet.*)

545/1

CHEMISTRY

Paper 1

April,2026

2 hours

For Scorers' Use Only

Total Weighted Score		
Initials		

CHEMISTRY DEPARTMENT EXAMINANTIONS BOARD

Uganda Certificate of Education

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of **two** *sections;* **A** *and* **B**.

It has **six** *examination items*.

Section **A** *has* **two compulsory** *items. Responses to section* **A must** *be written in the spaces provided.*

Section **B** *has* **two** *parts*; **I** *and* **II.** *Respond to only* **one** *item from* **each** *part.*

Responses to section **B must** *be written in the answer booklet*(*s*) *provided.*

Respond to **four** *items in all.*

Responses to each item in section **B** *should start on a fresh page.*

Any additional item(*s*) *responded to will* **not** *be scored.*

For Scorers' Use Only

Item No.	Basic Code	Weighted Score	Scorers' Initials
1	T		
	D		
	E		
2	C		
	X		
	I		
_____(3/4)	P		
	S		
	B		
_____(5/6)	N		
	M		
	Be		
Total Weighted Score			

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SECTION A

Respond to **all** *the items in this section in the spaces provided.*

Item 1

Hannah and Harriet are two sisters who live in an area where they use borehole water when washing clothes. As they were walking on a slippery and muddy road, they slipped and fell making their white dresses to become dirty.
Upon reaching home, Hannah decided to use a product **J** to clean her dress but the dress remained with brown spots, despite her rinsing it several times. Harriet decided to use product **K**, and her dress was fully cleaned after one rinse.
Hannah remained confused about the outcome of her cleaning process compared to that of her sister and she has reached out to you for help.

Task:

(a) Identify the categories of the products and their function.

_____

_____

_____

_____

_____

_____

_____

For Scorer's Use Only

Basis Code	**T**
Score	

(b) Explain the side effect(s) associated with the use of the products.

_____

_____

_____

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_____

_____

For Scorer's Use Only

Basis Code	**D**
Score	

(c) Guide Hannah and Harriet to make an evaluation of product **J** and product **K**.

_____

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_____

_____

_____

For Scorer's Use Only

Basis Code	**E**
Score	

Item 2

The fire brigade committee has opted to obtain a new fire extinguishing compound to be filled in fire extinguishers to be used to minimize several fire outbreaks in Kamba town. The committee reached out to your school for help and your chemistry teacher chose you to assist the committee.

The chemistry teacher told you that the best compound to be used is chemically written as $\mathbf{XY_2}$ (*where* **X** *and* **Y** *are* **not** *the actual symbols of the elements*). The teacher further told you that **X** belongs to group IV and period 2, while **Y** belongs to group VI and period 2 of the periodic table. Figure **1** shows a periodic table of the first 20 elements.

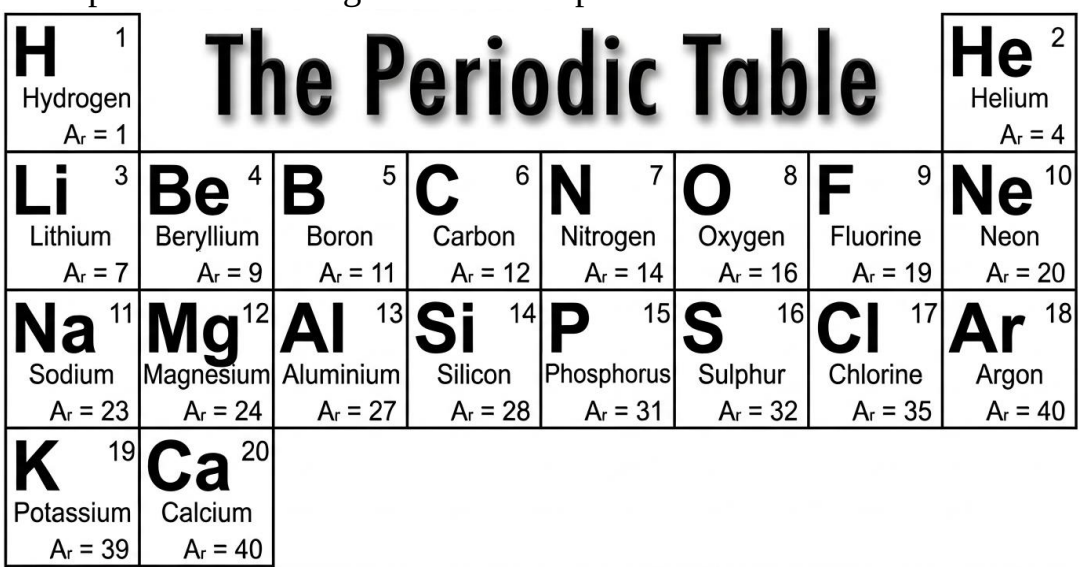

Fig.1. Periodic table of the first 20 elements

Task:

(a) Explain the nature of substances **X** and **Y**, and identify them.

(b) Describe how **X** and **Y** form the compound.

For Scorer's Use Only

Basis Code	**C**
Score	

(c) Explain the properties of the compound.

(d) Give any other use of the compound.

For Scorer's Use Only

Basis Code	**X**
Score	

(e) Give **one** impact of having a lot of compound $\mathbf{XY_2}$ in the environment and how it can be mitigated.

For Scorer's Use Only

Basis Code	**C**
Score	

SECTION B

Part I

Respond to only **one** *item in this part in the answer booklet*(*s*) *provided.*

Item 3

The iron sheets are part of the token which the government promised the reformed Karamojong warriors who voluntarily handed over their guns and embraced peace.
To avail them, a private investor has been cleared to set up a plant that will extract iron from iron ores in Kabale district.
A workshop involving students and community members was organized to create awareness to the issue.
However, the students and community members were concerned about the specific process leading to production of iron, the economic and social benefits of the manufacturing plant and how any side effect(s) is/are mitigated.
The government has chosen you to answer the concerns of students and community members during the workshop.

Task:
Write details of the key issues you will present during the workshop.

Item 4

The government has cleared a foreign crude oil development company to construct a crude oil refinery in Hoima district as shown in figure **2,** so as to increase on production of crude oil fractions used to run automobiles.

Fig. 2. Crude oil refinery

However, the residents are eager to know how the crude oil refining process will be conducted, the social economic benefits of the refinery and how the refinery/plant will impact their area.
The company CEO has selected a team of specialists in which you are among, to meet the residents and clarify about what they would like to know.

Task:
Make a write-up that you will present to residents upon meeting them.

Part II

Respond to only **one** *item from this part in the answer booklet(s) provided.*

Item 5

People of Kigogoma town have been found to be practicing zero grazing, cutting down trees and papyrus reeds from nearby forest and swamp for construction and making mats respectively as shown in figures **3, 4** and **5**. This has resulted into several challenges in the town.

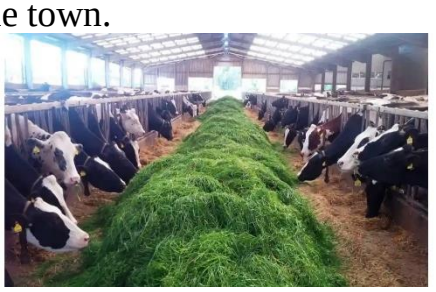

Fig. 3. Zero grazing

Fig. 4. Cut down trees

Fig. 5. Papyrus reeds

The town environment management authority has decided to start an environmental conservation campaign to sensitize residents about sustainable use of the affected natural resource.

You are a patriotic citizen who has decided to join the campaign.

Task:

Make a write-up you can use to sensitize fellow residents during the campaign.

Item 6

One of the most important sources of energy in industries is coal. It enhances industrial productivity and serves the community's needs. However, overdependence on coal has proven to be harmful to the environment hence a need to educate the community on mitigative measures. Figure **6** below shows coal mining

Fig. 6. Coal mining

You have been recommended by your school to be part of the sensitization team to educate the community.

Task:

Make a write-up you will use during sensitization.

END.